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SPECIAL TOPIC IN DATA ENGINEERING (SECP3843)

SECTION 01

GROUP 26

TECHNICAL REPORT

TOPIC 4:

ETL, ELT AND DBT

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EXECUTIVE SUMMARY

This report covers the history of data processing methods in data engineering, Extract, Transform, Load (ETL), Extract, Load, Transform (ELT) and the role of Data Build Tool (DBT) in the current data architecture. With increased use of data by organizations in making decisions, the shortcomings of the traditional ETL, especially with regard to scalability and processing speed, have become more pronounced with regard to processing large and complex datasets. ELT in its turn has become more dynamic, enabling data to be loaded into cloud-based data warehouses first and then transformed. In addition, DBT has been introduced as a key tool in modern data pipelines with supporting efficient data transformation. The report also examines practical uses and points out the advantages and difficulties of these methods. Although ELT and DBT have great scalability and productivity benefits, other concerns, including cost management and data governance, are still worth discussing. According to the results, it is suggested that organizations should implement a moderate approach that would utilize the newest data tools without sacrificing the good governance practices to guarantee efficiency and sustainability in the long-term.

1.0 INTRODUCTION

In the modern data driven world, efficient data processing is essential to organizations in order to aid decision making and business intelligence. The importance of data engineering has been on the rise to ensure that the raw data can be converted into meaningful insights. The ETL process has traditionally been popular to integrate and prepare data to be analyzed. Although ETL has been effective over many years, it frequently fails to scale and perform to the needs of a modern big data environment. As cloud computing have emerged, a more recent trend of ELT has become popular. As opposed to ETL, ELT moves the transformation process to the post-loading of a data warehouse and enables organizations to harness the computing capabilities of contemporary cloud systems. According to Stonebraker et al. (2018), current data architectures are being built to take advantage of the ability to scale storage and processing infrastructure, and ELT would be a more appropriate solution to big data applications. Along with this development, there are other tools that have come up like Data Build Tool (DBT). DBT allows data engineers and analysts to make transformations in the data warehouse via SQL. This enables more structured and sustainable data pipelines. A well-developed data transformation processes are the key to data consistency and reliability in contemporary data systems.

Consequently, the purpose of this report is to discuss the variations between ETL and ELT and the role of DBT in data engineering today. The discussion will dwell on the functionality of these methods, their benefits and weaknesses, and the role that they play in creating scalable and efficient data architectures in modern enterprise settings.

2.0 BACKGROUND / LITERATURE REVIEW

Traditionally, data integration relied on Extract, Transform, Load (ETL) process. By converting data stored in centralized servers prior to loading it into a warehouse, the organizations guaranteed data that was highly structured data reached business intelligence layers (Kimball and Ross, 2013). Nevertheless, with the explosion of global data, the strict preprocessing bottlenecks of the traditional ETL were unable to perform well at scale, and the elasticity of big data needs was required (Nawaz, 2026; Zaharia et al., 2016).

To bypass these limits, cloud computing led to a paradigm shift of Extract, Load, Transform (ELT). ELT loads information to a cloud data warehouse and processes it within the system. By taking advantage of the distributed compute and decoupled storage of cloud-native platforms, organizations ingest huge amounts of raw data in real time and transform it directly in the cloud. Platforms like Snowflake and Google BigQuery distribute computing resources independently of storage to manage large datasets. This system increases scalability and flexibility, enabling the cost-effective storage of raw data and transforming it only when it is used in the analytical workloads (Armbrust et al., 2021; Dageville et al., 2016).

ELT resolved the infrastructure issues, yet it established a new research deficiency: the need to handle complicated changes within the warehouse tended to cause poor data management. To address these concerns, dbt (data build tool) became the standard of the industry in terms of coordination (Murugan, 2025). It is an SQL transformation in a warehouse, constructed by dbt Labs. Through software engineering concepts such as modular SQL, automated testing, and Git-based version control directly to the transformation layer, tools such as dbt transformed data transformation into a highly structured and robust process (Reis and Housley, 2022). Finally, the combination of scalable ELT architectures and dbt creates the Modern Data Stack, but further studies are needed to learn how large companies can fully utilize these solutions to ensure perfect data quality.

3.0 ANALYSIS AND DISCUSSION

Recent runtime studies indicate that ELT is much faster than the conventional ETL, and it uses the parallelization capabilities of cloud computing (Wang et al., 2023).

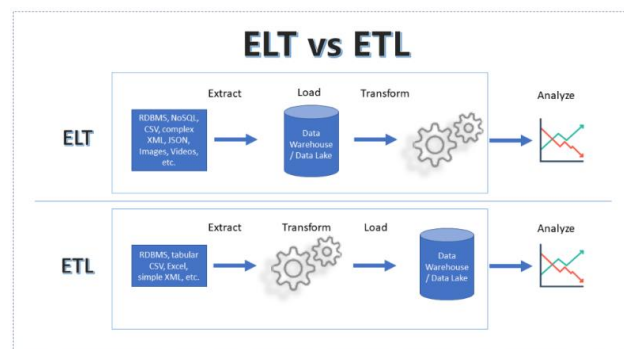


Figure 1: Architectural comparison of flow of traditional ETL and modern ELT using dbt orchestration.

3.1 TECHNOLOGY ARCHITECTURE AND FUNCTIONALITY

The current data architecture show that storage and compute run independently of each other. This enables the ELT sequence, which works with data in cloud architecture. The fundamental technology is divided into four layers:

- Ingestion Layer (Extract & Load): Fivetran or Airbyte are tools that transfer raw data (SaaS, CRMs, ERPs) into the cloud warehouse without first changing them.
- Storage Layer (The Warehouse): Snowflake or Google BigQuery are the vendors of the single source of truth that stores petabytes of raw data at a low cost (Dageville et al., 2016).
- Transformation Layer (dbt): It has modular SQL and Jinja templating to convert raw data into modeled clean tables. It brings software engineering, such as version control and CI/CD (Reis & Housley, 2022).
- Observation & Governance Layer: Data quality and adherence to regulations such as GDPR or HIPAA is guaranteed through automated testing and metadata tracking.

3.2 ENTERPRISE CASE STUDIES AND APPLICATIONS

The productivity effect of changing ETL to ELT + dbt can be estimated in different fields of industry:

Financial Services & Compliance: McKinsey & Company (2023) indicate that this architecture is the key to next-gen banking success because it can manage the 50 percent growth in data variety observed in digital banking. Although traditional ETL is still applied to mask PII, the move to ELT of non-sensitive data has eliminated days of report latency and decreased it to minutes.

Media & streaming (The Netflix Model): Netflix consumes billions of rows of clickstream data on a daily basis using ELT. This enables data scientists to work with raw logs that can be used with machine learning and analytics engineers to create structured dashboards to business executives at the same time (Wang et al., 2023).

3.3 STRATEGIC ADVANTAGES AND BUSINESS IMPACT

ELT is based on the unlimited nature of the compute power provided by the cloud as opposed to legacy ETL that is limited to the physical RAM of a transformation server. According to research conducted by the IEEE (2023), ELT can be utilized in runtime optimization that can enhance the data processing throughput by 3.5x compared to conventional approaches.

Dbt enables SQL-capable analysts. This shift results in the formation of the new category of professionals the so-called Analytics Engineer a professional that is knowledgeable on both business logic and code-based best practices.

3.4 ANALYSIS OF CHALLENGES AND STRATEGIC POSITIONING

- Strong points: Fast deployment cycles; Git based version control; high visibility due to automated documentation.
- Weaknesses: Cost Volatility. According to Gartner (2024), inefficient dbt models may add to the cost of cloud compute by 25-40 percent when they are not transferred.
- Opportunities: Integrating data in real-time; SQL production with AI; scaling the "Data Mesh" to decentralized business units.
- Threats: Shadow IT & Data Swamps. Decentralization change may result in the emergence of conflicting business metrics definitions (Harvard Business Review, 2023).

3.5 FUTURE TRENDS AND STRATEGIC IMPLICATIONS

The future of AI-dbt is integration. As per Harvard Business Review (2022), real-time streaming analytics is the next outstanding digital advantage. Moreover, the dbt IDE is being supported by Generative AI to implicitly propose SQL optimization, produce documentation, and save users up to 30 more percent of time on manual coding by 2026.



Figure 2: Architectural of DBT Snowflake

4.0 RECOMMENDATION

In choosing a suitable data processing architecture, organizations must balance the strategy with the technical considerations and long-term business objectives. Instead of providing a blanket solution, a hybrid approach may be a viable option in many cases to achieve the best outcome. As an example, a traditional ETL can still be the appropriate choice in case of sensitive or highly regulated information where strong control of the transformation is needed. Nevertheless, Armbrust et al. (2021) discuss organizations that work with large-scale and high-growing datasets, it is typically more efficient to adopt an ELT-based architecture as it can make use of cloud scalability and parallel processing.

Moreover, it is recommended that organizations use DBT as a component of their current data stack to standardize and simplify transformation processes. Such an approach, as Reis and Housley (2022) emphasize, can not only improve maintainability but also facilitate the process of collaboration between the technical and business teams, which improves the speed with which the insights can be delivered. Meanwhile, organizations should be mindful of the operational issues related to modern data architectures. Specifically, cost management and governance of cloud-based ELT environments should be done carefully to avoid inefficiency. Thus, a balanced approach that integrates current technologies with solid governance frameworks will empower organizations to achieve maximum positive outcomes of scalable and efficient data engineering systems.

5.0 CONCLUSION

Generally speaking, the data engineering practice development trend represents a wider change to more flexible, scalable and cloud-based solutions. Although previous models offered a solid basis of performing structured data processing, the current models are increasingly prepared to tackle the complexity and volume of the data environment today. Moreover, the introduction of transformation tools like DBT proves the increasing role of data workflows as shaped by the concepts of software engineering. This not only enhances the quality and consistency of data but also allows working of teams more efficiently. According to Stonebraker et al. (2018), contemporary data systems are constantly developing due to the increasing requirements of speed, scalability, and analytical power.

In the future, data engineering will probably be influenced by further innovations in cloud computing, automation, and real-time data processing. Those organizations capable of strategically adopting and managing these technologies will be in a better position to create timely insights and be at an advantage of competition. This means that knowledge and successful implementation of current data engineering strategies will continue to play a key role in the development of sustainable and future-proof data infrastructures.

6.0 REFERENCES

- Armbrust, M., Ghodsi, A., Xin, R., & Zaharia, M. (2021). Lakehouse: A new generation of open platforms that unify data warehousing and advanced analytics. *Proceedings of the 11th Annual Conference on Innovative Data Systems Research (CIDR)*.
https://cidrdb.org/cidr2021/papers/cidr2021_paper17.pdf
- Chambers, B., & Zaharia, M. (2018). *Spark: The definitive guide*. O'Reilly Media.
<https://www.oreilly.com/library/view/spark-the-definitive/9781491912201/>
- Dageville, B., Cruanes, T., Zukowski, M., Antonov, V., Avanes, A., ... Bica, A. (2016). The Snowflake elastic data warehouse. *Proceedings of the 2016 International Conference on Management of Data*, 215–226. <https://doi.org/10.1145/2882903.2903741>

- Gartner. (2024). *Essential skills to build modern data engineering systems*. Gartner Data & Analytics Research. <https://www.gartner.com/en/documents/4015694>
- Harvard Business Review. (2022). The next great digital advantage. *Harvard Business Review Magazine*, May-June Issue. <https://hbr.org/2022/05/the-next-great-digital-advantage>
- Harvard Business Review. (2023). *From operational data maintenance to strategic data architecture: Master data management at Chr. Hansen*. HBR Case Studies. <https://hbr.org/store/case-studies>
- Kimball, R., & Ross, M. (2013). *The data warehouse toolkit: The definitive guide to dimensional modeling* (3rd ed.). Wiley. <https://www.kimballgroup.com/data-warehouse-business-intelligence-resources/books/data-warehouse-dw-toolkit/>
- McKinsey & Company. (2023, February 28). *Next-gen banking success starts with the right data architecture*. McKinsey Technology Insights. <https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/tech-forward/next-gen-banking-success-starts-with-the-right-data-architecture>
- Murugan, A. S., Saravanan, S. K., Nidhya, G. S., Sasikala, K., Pandey, P., & Arasan, E. (2025). Leveraging Data Build Tool for efficient data management in data center architectures. *2025 11th International Conference on Communication and Signal Processing (ICCSP)*, 502-507. IEEE. <https://ieeexplore.ieee.org/abstract/document/11089387>
- Nawaz, M. K. (2026). *Exploring end-to-end data engineering: A GCP case study*. Theseus Academic Repository. <https://www.theseus.fi/handle/10024/858907>
- Reis, J., & Housley, M. (2022). *Fundamentals of data engineering: Plan and build robust data systems*. O'Reilly Media. <https://www.oreilly.com/library/view/fundamentals-of-data/9781098108298/>
- Stonebraker, M., Madden, S., Abadi, D. J., Harizopoulos, S., Hachem, N. I., & Helland, P. (2018). The end of an architectural era: It's time for a complete rewrite. *Communications of the ACM*, 463–489. <https://doi.org/10.1145/3226595.3226637>
- Wang, L., Zhang, Y., & Chen, X. (2023). Extract-Load-Transform (ELT) process runtime analysis and optimization. *2023 IEEE 5th International Youth Conference on Radio Electronics, Electrical and Power Engineering (REEPE)*, 34-40. <https://doi.org/10.1109/REEPE57272.2023.10140280>

Zaharia, M., Chen, A., Davidson, A., Ghodsi, A., Hong, S., Konwinski, A., ... Parkhe, M. (2016).
 Apache Spark: A unified engine for big data processing. *Communications of the ACM*, 59(11),
 56–65. <https://doi.org/10.1145/2934664>

7.0 LOGBOOK

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Member 2 Name: Ling Yu Qian

Project Title: ETL, ELT AND DBT

Date & Time	Task & Tools Used	Keywords/Prompts
5/4/2026 9:00am	Google Scholar	ETL vs ELT cloud data architecture
5/4/2026 10:13am	Google Scholar	Modern data engineering trend
5/4/2026 03:00pm	ChatGPT	Validate the introduction, is the explanation enough?
6/4/2026 10:00am	Google Scholar	Evolution of ETL to ELT in cloud architectures 2021-2025
6/4/2026 02:30pm	Gemini 3.1 pro	Give me credible IEEE and HBR references related to ELT and dbt for academic learning purposes
6/4/2026 05:00pm	Gemini 3.1 pro	Analyze my Background and Discussion section: Does it successfully synthesize Reis & Housley (2022) and Wang et al. (2023)?
8/4/2026 23:30pm	ChatGPT	Check if there is too much repetition in the report and grammar check.